PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Electrochemistry (Question Bank # 06)

MM : 71 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. Na - Amalgam is prepared by electrolysis of aq. NaCl solution using 10 gm Hg cathode. How many Faraday's of electricity is required to prepare 18.7% Na-Amalgam. (current Eff. - 50%)

(A) 0.081 F (B) 0.16 F

(C) 0.2 F (D) 0.1 F

2. Molar conductances of $BaCl_2$, H_2SO_4 and HCl are x_1, x_2 and x_3 $Scm^2\ mol^{-1}$ at infinite dilution. If specific conductance of saturated $BaSO_4$ solution is of y Scm^{-1}, then k_{sp} of $BaSO_4$ is

(A) $\dfrac{10^3 y}{2(x_1+x_2-2x_3)}$

(B) $\dfrac{10^6 y^2}{(x_1+x_2-2x_3)^2}$

(C) $\dfrac{10^6 y^2}{4(x_1+x_2-2x_3)^2}$

(D) $\dfrac{x_1+x_2-2x_3}{10^6 y^2}$

3. The conductivity of a saturated solution of Ag_3PO_4 is 9×10^{-6} S m^{-1} and its equivalent conductivity is 1.50×10^{-4} S m^2 $equivalent^{-1}$. The K_{sp} of Ag_3PO_4 is

(A) 4.32×10^{-18} (B) 1.8×10^{-9}

(C) 8.64×10^{-13} (D) None of these

4. What would be the reduction potential of an electrode at 298 K, which originally contained 1 M $K_2Cr_2O_7$ solution in acidic buffer solution of pH = 1.0 and which was treated with 50% of the Sn necessary to reduce all $Cr_2O_7^{2-}$ to Cr^{3+}. Assume pH of solution remains constant.

Given: $E^\circ_{Cr_2O_7^{2-}/Cr^{3+},H^+} = 1.33$ V; log 2 = 0.3 ; $\dfrac{2.303RT}{F} = 0.06$

(A) 1.285 V (B) 1.193 V

(C) 1.187 V (D) 1.473 V

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5. At certain temperature (T) if conductivity of pure water is 5.5×10^{-7} S cm^{-1} then calculate pOH of pure water at temperature T.

Given: $\lambda^{\infty}_{H^+} = 350$ S cm^2 eq^{-1}; $\lambda^{\infty}_{OH^-} = 200$ S cm^2 eq^{-1}

(A) 5 (B) 6 (C) 7 (D) 8

6. In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is

$$CH_3OH(\ell) + \frac{3}{2}O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$$

At 298 K standard Gibb's energies of formation for $CH_3OH(\ell)$, $H_2O(\ell)$ and $CO_2(g)$ are –166.2, –237.2 and –394.4 kJ mol^{-1} respectively. If standard enthalpy of combustion of methanol is –726 kJ mol^{-1}, efficiency of the fuel cell will be

(A) 90% (B) 97%

(C) 80% (D) 87%

(ii) One or more options correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

7. Select the correct statements-

(A) Generally oxidation potential of anodic half cell decreases during reaction

(B) If standard electrode potential of an electrode is greater than zero then it's reduced form is more stable compared to hydrogen gas

(C) $E^{\circ}_{Oxd.(H_2O_2/O_2)} = -E^{\circ}_{Red.(H_2O_2/H_2O)}$

(D) Inert electrodes provide their surface for oxidation or reduction reactions.

8. The cell potential for the unbalanced chemical reaction

$$Hg_2^{2+} + NO_3^- + H^+ \longrightarrow Hg^{2+} + HNO_2 + H_2O \quad ; \quad E^{\circ}_{cell} = 0.02 \text{ V}$$

is measured under standard state conditions in the electrochemical cell. $\left(\frac{2.303\,RT}{F} = 0.06\right)$

Then which is/are the correct statement(s) :-

(A) Solution in cathodic compartment is acidic

(B) Solution in Anodic compartment is acidic

(C) Cell potential will be zero at pH = $\frac{2}{9}$, if activity of other components are equal to one.

(D) 0.6 moles of electron pass through the circuit when 0.6 moles of Hg^{2+} are produced in the cell

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9. Pick out the **correct** statements among the following from inspection of standard reduction potentials (Assume standard state conditions).

Cl_2 (g) + 2e	$\rightleftharpoons$	$2Cl^-$(aq.)	$E^o_{Cl_2/Cl^-}$ = + 1.36 volt
Br_2 (*l*) + 2e	$\rightleftharpoons$	$2Br^-$ (aq.)	$E^o_{Br_2/Br^-}$ = + 1.09 volt
I_2(s) + 2e	$\rightleftharpoons$	$2I^-$ (aq.)	$E^o_{I_2/I^-}$ = + 0.54 volt
$S_2O_8^{2-}$ (aq.) + 2e	$\rightleftharpoons$	$2SO_4^{2-}$ (aq.)	$E^o_{S_2O_8^{2-}/SO_4^{2-}}$ = + 2.00 volt

(A) Cl_2 can oxidise SO_4^{2-} from solution

(B) Cl_2 can oxidise Br^- and I^- from aqueous solution

(C) $S_2O_8^{2-}$ can oxidise Cl^-, Br^- and I^- from aqueous solution

(D) $S_2O_8^{2-}$ is added slowly, Br^- can be reduce in presence of Cl^-

10 Indicate the correct statements -

(A) Conductivity cells have cell constant values independent of solution filled into the cell

(B) DC (direct current) is not used for measuring the resistance of a solution

(C) Kohlrausch law is valid both for strong and weak electrolytes

(D) The κ decreases but λ_m and $\lambda_{eq.}$ increase on dilution

(iii) Paragraph Type

This section contains **1 paragraph** , describing theory, experiment, data etc. **3 questions** relate to one paragraph with 3 questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for question no. 11 to 13

In quantitative estimations of any element using an oxidizing agent it is essential to predict which substance (s) gets oxidized by the oxidizing agent. In case more than one substance are getting oxidized then the oxidizing agents gets distributed in all the reactions taking place. From this information and the data given below answer the questions that follow.

$Fe^{+2} \rightarrow Fe^{+3} + e^-$	$E^o_{O.P.}$ = – 0.77V
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	$E^o_{R.P.}$ = 1.51 V
$2Cl^- \rightarrow Cl_2 + 2e^-$	$E^o_{O.P.}$ = – 1.36V
$2SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$	$E^o_{O.P.}$ = – 2.0V
$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$	$E^o_{O.P.}$ = – 0.77V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	$E^o_{R.P.}$ = 1.33V

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11. Which of the following can not be used for quantitative estimation of FeC_2O_4?

(A) $K_2Cr_2O_7$ in presence of dil H_2SO_4.

(B) $K_2Cr_2O_7$ in presence of dil. HCl

(C) $KMnO_4$ in presence of dil. HCl.

(D) $KMnO_4$ in presence of dil. H_2SO_4

12. Millimoles of FeC_2O_4 in the solution if 50 ml of 0.1 M $K_2Cr_2O_7$ is used for its oxidation in presence of dil. H_2SO_4.

(A) 10 millimoles

(B) 30 millimoles

(C) 20 millimoles

(D) quantitative estimation can't be made from given data

13. Millimoles of FeC_2O_4 in the solution, if 50 ml of 0.1 M $KMnO_4$ is used for its oxidation in presence of dill. HCl if 3.5 millimoles of Cl_2 is obtained along with other products.

(A) 25/3 millimoles

(B) 6 millimoles

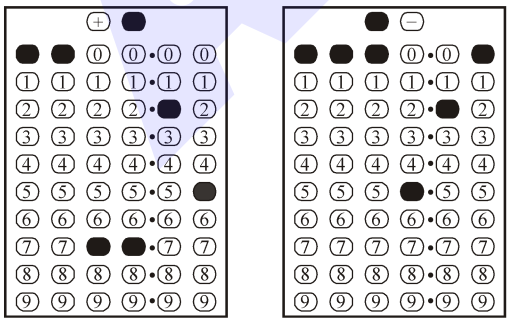
(C) 12.5 millimoles

(D) quantitative estimation can't be made from given data

SECTION-II: (Integer Second Decimal)

- This section contains **5** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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1. Calculate the magnitude of electrode potential in volt having reaction: $Ag_2S(s) + 2e^- \rightleftharpoons 2Ag(s) + S^{2-}(aq)$. in a solution buffered at pH = 3 and which is also saturated with 0.1 M H_2S (aq).

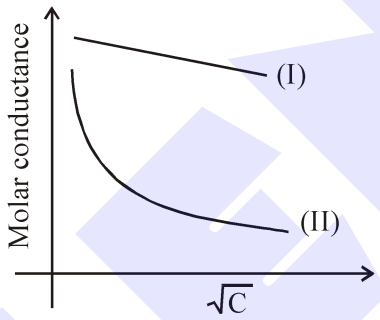
Given For H_2S : $K_{a_1} = 10^{-7}$ and $K_{a_2} = 10^{-14}$, $K_{sp}(Ag_2S) = 10^{-49}$, $E^\circ_{Ag^+/Ag} = 0.8$.

Use $2.303 \times \frac{RT}{F} = 0.06$

2. A current of 0.1 amp is passed though excess of molten mixture of Al_2O_3 and Na_3AlF_6 for 96.5 hours.

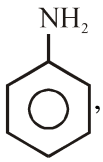
The mass of Al (in gm) deposited at the cathode, with $\frac{1000}{10.8}\%$ current efficiency is (Al = 27).

3. Given graph was plotted between the molar conductance of various electrolytes and $\sqrt{C}$ generally (I) & (II) types of graphs are obtained.



Among the following the number of species having similar graph to the (II) is

CH_3COOH, 

, KCl, HCl, 

,

$AgNO_3$, CH_3COONa, Na_2HPO_3, H_3PO_4, 

4. For an element 'X' the process of oxidation is :

$X_2O_4^{-2} \longrightarrow$ New compound

If 965 A current when passed for 100 seconds discharged 0.1 mol of $X_2O_4^{-2}$, find oxidation state of X in new compound ?

5. Calculate e.m.f of cell

$Ag \mid AgNO_3$ (0.1M) $\parallel Ag(NH_3)_2^+$ (0.01 M), NH_3(1M) $\mid Ag$

Given : - $K_f([Ag(NH_3)_2^+]) = 10^8$

$\frac{2.303RT}{F} = 0.06$

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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the following Column–I and Column–II :

Column–I	**Column–II**
(A) Metal–metal ion half cell	(P) $Cl^- \mid AgCl \mid Ag$
(B) Gas–gas ion half cell	(Q) $Pt \mid Cr^{2+}(C_1M), Cr^{3+}(C_2M)$
(C) Redox half cell	(R) $Pt \mid O_2 \mid OH^-$
(D) Metal insoluble metal salt-anion half cell	(S) $Sn^{2+} \mid Sn$
	(T) $Pt \mid Sn^{4+}, Sn^{2+}$

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ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1.	Ans. (C)	2.	Ans. (B)	3.	Ans. (A)	4.	Ans. (C)
5.	Ans. (B)	6.	Ans. (B)				

(ii) One or more options correct Type : 4(–1)

7.	Ans. (A,B,D)	8.	Ans. (A,C,D)	9.	Ans. (B,C)	10.	Ans. (A,B,C,D)

(iii) Paragraph Type : 3(0)

11.	Ans. (C)	12.	Ans. (A)	13.	Ans. (B)

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans. 0.19	2.	Ans. (3.00)	3	Ans. (4.00)	4.	Ans. (8.00)
5.	Ans. (0.54)						

SECTION–IV : Matrix-Match Type: 8(0)

1. Ans. (A)–(S) ; (B)–(R) ; (C)–(Q) (T) ; (D)–(P)

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